Topic 4: Saving, Investment & the Financial System

Introduction to Economics II

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Based on Mankiw & Taylor, Chapter 23

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Where We Are: From Production to Allocation

Recall (Topics 1-3): In Topic 3, we answered: **What determines how much an economy produces?**

$$\bar{Y} = AF(L, K, H, N)$$

We established three key ideas:

- Output is determined by **supply**: factor endowments and technology
- The classical dichotomy holds: nominal variables don't affect real output
- Capital (K) is crucial for productivity and growth

But capital depreciates over time. To maintain and grow K, the economy needs **investment** (I).

K is a **stock** — fixed at any point in time (Topic 3).
I is a **flow** — the amount of output devoted each period to new capital.
Today's investment determines tomorrow's stock: $\Delta K = I - \delta K$

Where do the resources for investment come from?

The Question of Topic 4

Given that $\bar{Y}$ is already determined, how is output allocated?

Every unit of output must go somewhere:

- Some is **consumed** by households today
- Some is **used by government** for public services
- Some is **invested** to build capital for tomorrow

This is a question about composition, not level:

- We are **not** asking "how much is produced" (Topic 3 answered that)
- We **are** asking "of what is produced, how much goes to each use?"

Central insight: There is a **trade-off** between consuming today and investing for the future. The **real interest rate** (r) coordinates this.

Roadmap of Topic 4

1. **Allocation of Output:** Where does production go?
   - Private saving, public saving, national saving
   - The identity: $S = I$
2. **The Financial System:** How do funds flow from savers to investors?
   - Bonds, stocks, banks
3. **The Loanable Funds Market:** What coordinates saving and investment?
   - Supply, demand, the role of r
4. **Formalisation:** Algebra behind the curves
   - Consumption and investment functions, solving for r^*
5. **Fiscal Policy:** How does government affect this allocation?
   - Budget deficits, crowding out, and the role of elasticities

How Is Output Allocated?

We are still working in a closed economy (no international trade):

Recall (Topics 1-3): From Topic 2, every unit of output is either consumed, invested, or used by government:

$$Y = C + I + G$$

Important: This is an **accounting identity**, not a theory.

It tells us that output has three possible **uses**.

In the long run, $Y = \bar{Y}$ is fixed by supply (Topic 3).

The question is how $\bar{Y}$ is **divided** among C, I, and G.

We will add the open economy (NX) in Topic 6.

Disposable Income: What Households Receive

Households earn income Y, but they don't keep all of it:

- They pay **taxes (T)** to government
- What remains is **disposable income**

Disposable Income: $Y^D = Y - T$

The income households have available to use

Households allocate their disposable income to two uses:

1. **Consumption (C):** Buying goods and services *today*
2. **Private Saving ($S_{private}$):** Setting aside resources for the *future*

$$Y - T = C + S_{private}$$

Private Saving

Private Saving: $S_{private} = (Y - T) - C$

The portion of disposable income that households **do not** consume

Example:

- María earns €30,000/year
- Pays €6,000 in taxes
- Consumes €20,000

$$\begin{aligned} &\text{Disposable income:} && 30{,}000 - 6{,}000 = 24{,}000 \\ &\text{Private saving:} && 24{,}000 - 20{,}000 = 4{,}000 \end{aligned}$$

Key point: Every euro not consumed is a euro saved — a euro potentially available for investment.

Public Saving: The Government Budget Balance

The government also allocates resources:

Public Saving: $S_{public} = T - G$

Tax revenue minus government spending

Three scenarios:

1. **Budget Surplus:** $T > G \Rightarrow S_{public} > 0$
   - Government collects more than it spends — adds to national saving
2. **Budget Deficit:** $T < G \Rightarrow S_{public} < 0$
   - Government spends more than it collects — subtracts from national saving
3. **Balanced Budget:** $T = G \Rightarrow S_{public} = 0$

National Saving

Combining private and public saving:

$$\textbf{National Saving: } S = S_{private} + S_{public}$$
$$S = (Y - T - C) + (T - G)$$
$$S = Y - C - G$$

Interpretation:

- National saving is the part of output **not absorbed** by consumption or government
- It measures how much output is "left over" for building future capital

Example:

- If $\bar{Y} = 100$, $C = 60$, $G = 20$
- Then $S = 100 - 60 - 20 = 20$
- 20% of output is available for investment

The Fundamental Identity: $S = I$

Starting from the allocation identity:

$$\begin{aligned} Y &= C + I + G \\ Y - C - G &= I \\ S &= I \end{aligned}$$

Fundamental Identity (Closed Economy):

$$S = I$$

National Saving = Investment

This is an <u>accounting identity</u> — it holds by definition:

- The resources not consumed *must* go to investment
- But *who* saves and *who* invests are different agents
- How are they coordinated? → The financial system

Two Views of the Same Equilibrium

What we have just derived can be read in two equivalent ways:

View 1: OA = DA

$$\underbrace{\bar{Y}}_{\text{Agg. Supply}} = \underbrace{C + I + G}_{\text{Agg. Demand}}$$

Total output = Total spending

$\Longleftrightarrow$

View 2: S = I

$$\underbrace{\bar{Y} - C - G}_{S} = I$$

Resources not consumed = Resources invested

These are <u>not</u> two different equilibria — they are the same condition rearranged.

- In the **long run**, $\bar{Y}$ is fixed (Topic 3). Both views describe how that fixed output is **allocated**.
- The variable that ensures this equilibrium holds is the **real interest rate (r)**.
- We use the $S = I$ view — the **loanable funds market** — because it directly shows how r coordinates the allocation.

Numerical Example: Understanding $S = I$

An economy with the following values (in billions):

Variable	**Value**
Output ($\bar{Y}$)	€1,000
Consumption (C)	€600
Government spending (G)	€250
Taxes (T)	€200

Calculate:

- Investment: $I = \bar{Y} - C - G = 1{,}000 - 600 - 250 =$ **€150 bn**
- Private saving: $S_{priv} = (Y - T) - C = 800 - 600 =$ **€200 bn**
- Public saving: $S_{pub} = T - G = 200 - 250 =$ **-€50 bn** (deficit!)
- National saving: $S = 200 + (-50) =$ **€150 bn**

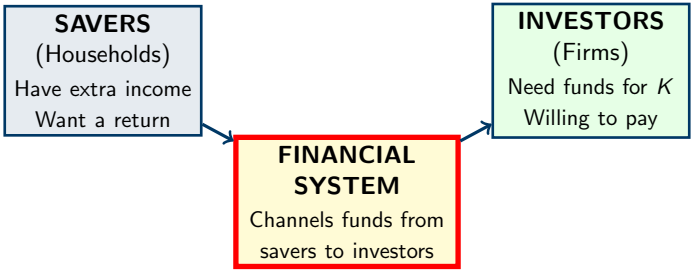
Verify: $S = I \Rightarrow 150 = 150$ ✓

The Problem: Matching Savers and Investors

$S = I$ must hold, but HOW does it happen in practice?

The challenge:

- Those who **save** (households) are not the same as those who **invest** (firms)
- María saves €4,000, but she doesn't build a factory
- A firm needs €1 million for equipment, but doesn't have it



Financial Markets: Bonds and Stocks

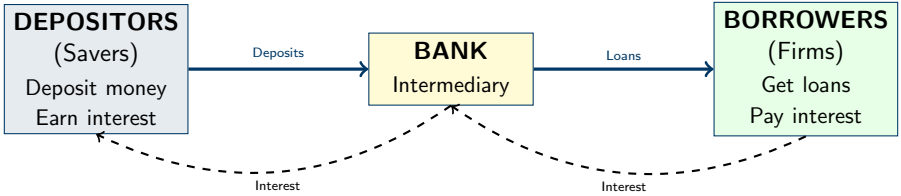
Savers can lend directly to firms through financial markets:

Bond Market	**Stock Market**
A **bond** is a certificate of debt	A **stock** (share) is partial ownership
Firm borrows €1,000 Promises to repay with interest	Firm sells ownership stakes Buyer gets share of future profits
Fixed return, must be repaid Lower risk for the saver	**Variable return**, no repayment Higher risk, higher potential reward
Example: Spanish government 10-year bond at 3%	*Example:* Buying Telefónica shares on the stock exchange

Both serve the same function: channel savings from households to firms that need funds for investment.

Financial Intermediaries: Banks

Not all lending is direct. Banks act as intermediaries:



How banks work:

- Pay depositors a low rate (e.g., 1%)
- Charge borrowers a higher rate (e.g., 5%)
- The difference (spread) covers costs and profits

Other intermediaries include investment funds and insurance companies, which pool savings from many individuals and invest in diversified portfolios of bonds and stocks.

Summary: The Financial System

All financial institutions serve the same fundamental purpose:

Channel resources from those who SAVE to those who INVEST

Ensuring that $S = I$ is achieved in practice

Institution	Savers get	Firms get	Type
Bond market	Interest payments	Direct funding	Market
Stock market	Dividends + gains	Equity capital	Market
Banks	Deposit interest	Loans	Intermediary

But at what "price" does this exchange happen?

→ We need a model: the **loanable funds market**, where the price is r

The Loanable Funds Market

A model for the allocation of output over time

The Loanable Funds Market

A single market where savers supply funds
and borrowers (investors) demand funds

Price: The real interest rate (r)

Key assumptions (simplifications):

- There is a single market for all saving and borrowing
- One interest rate clears the market
- The real interest rate (r) is what matters for decisions

What does r represent?

- For **savers**: the reward for postponing consumption
- For **investors**: the cost of borrowing to buy capital

Real vs Nominal Interest Rate

Why do we use the <u>real</u> interest rate?

Nominal Interest Rate (i)	**Real Interest Rate (r)**
The rate you see at the bank Example: bank offers 5%/year €100 today $\rightarrow$ €105 next year	Adjusted for inflation $r = i - \pi$ Measures change in *purchasing power*

Example:

- You deposit €100 at 5% nominal interest
- Inflation is 3%
- Next year you have €105, but prices rose 3%
- Real return: $r = 5\% - 3\% = 2\%$
- Your *real* purchasing power increased only 2%

Recall (Topics 1-3): From Topic 2: we measure inflation with the CPI or the GDP deflator.

From now on, "r" always means the real interest rate.

Supply of Loanable Funds: National Saving

Who supplies loanable funds? SAVERS

Supply of loanable funds = National Saving (S)

$$S = S_{private} + S_{public} = Y - C - G$$

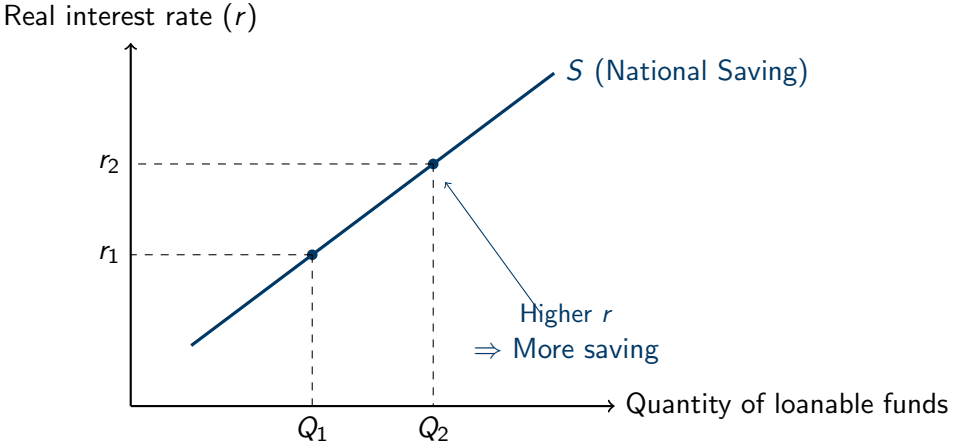
Why does supply slope upward?

- Higher $r \Rightarrow$ Saving becomes more attractive
- If the reward for postponing consumption is 10% instead of 2%, more people choose to save

What shifts the supply curve?

- Changes in taxes (T) or government spending (G)
- Changes in household preferences (thrift)
- Tax policies that affect saving incentives

Supply of Loanable Funds — Diagram



Key insight: The supply curve slopes upward because a higher interest rate makes saving more rewarding — more people choose to postpone consumption.

Demand for Loanable Funds: Investment

Who demands loanable funds? **FIRMS** (for investment in capital)

Demand for loanable funds = Investment (I)

Firms borrow to buy capital goods (machines, equipment, buildings)

Why does demand slope downward?
Firms invest only when the **expected return** of a project exceeds its **cost of financing** (r).

Example: A firm considers buying a machine for €100,000.

- Expected annual profit from the machine: €8,000 (8% return)
- If $r = 5\% \Rightarrow$ Cost of funds = €5,000 $\Rightarrow$ **INVEST** ✓
- If $r = 10\% \Rightarrow$ Cost of funds = €10,000 $\Rightarrow$ **DON'T INVEST** ×

Higher $r \Rightarrow$ fewer profitable projects $\Rightarrow$ less investment

Investment Projects and the Interest Rate

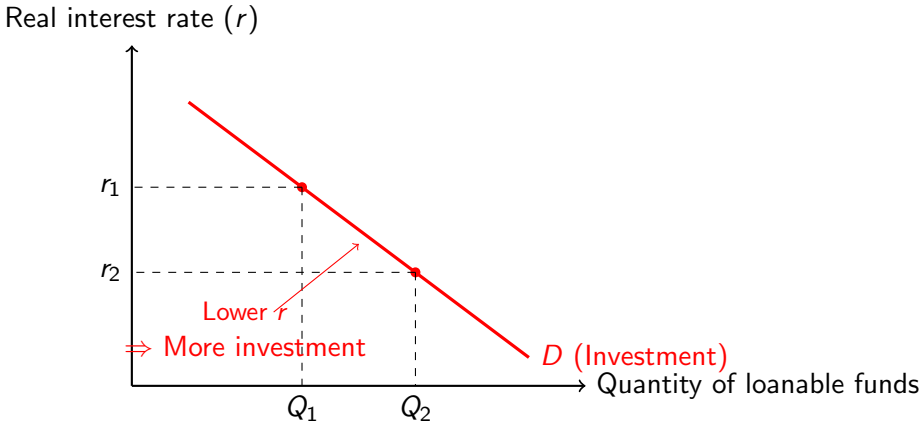
Think of the economy as having many possible investment projects, each with a different expected return:

Project	Cost	Expected return
New factory	€5M	12%
IT upgrade	€2M	9%
Office expansion	€3M	7%
Warehouse renovation	€1M	4%
Decorative lobby	€0.5M	2%

- If $r = 10\%$: only the factory is funded (€5M invested)
- If $r = 6\%$: factory + IT + office (€10M invested)
- If $r = 1\%$: all projects funded (€11.5M invested)

This is why the demand curve slopes down: as r falls, more projects become viable.

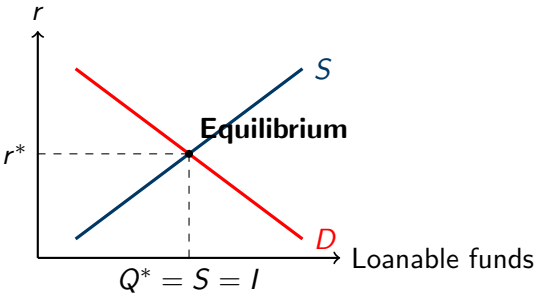
Demand for Loanable Funds — Diagram



Key insight: The demand curve slopes downward because higher interest rates make borrowing more expensive — fewer investment projects are profitable.

Equilibrium in the Loanable Funds Market

The interest rate adjusts to balance saving and investment



At the equilibrium interest rate r^*:

- Quantity of saving = Quantity of investment: $S = I$
- The interest rate is the "price" that coordinates the intertemporal allocation of output

How Does the Market Reach Equilibrium?

What if r is not at equilibrium?

Scenario 1: r too high

- Saving $>$ Investment (surplus of funds)
- Lenders compete for borrowers $\Rightarrow$ they lower the rate they charge
- r falls until $S = I$

Scenario 2: r too low

- Investment $>$ Saving (shortage of funds)
- Borrowers compete for scarce funds $\Rightarrow$ they accept higher rates
- r rises until $S = I$

The real interest rate adjusts to equilibrate the supply and demand for loanable funds.

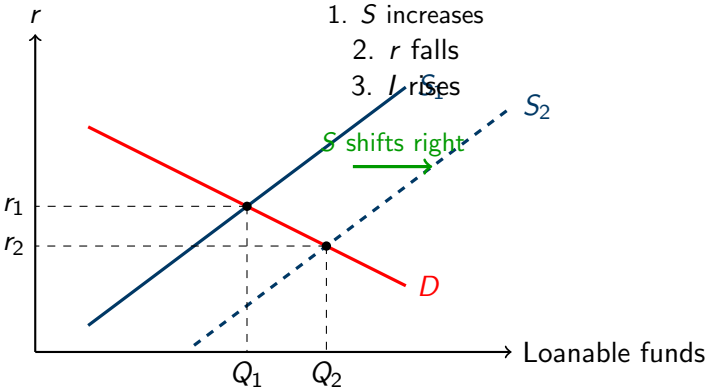
This is how the financial system coordinates the allocation of output.

What Shifts the Supply Curve?

Recall: Supply = National Saving = $S = Y - C - G$

1. **Tax incentives for saving**
   - Tax breaks on interest income make saving more attractive
   - Households save more at every $r \Rightarrow$ Supply shifts **RIGHT**
2. **Change in preferences (increased thrift)**
   - Households decide to consume less and save more
   - Supply shifts **RIGHT**
3. **Government budget deficit ($G \uparrow$ or $T \downarrow$)**
   - $S_{public} \downarrow \Rightarrow$ National saving $S \downarrow$
   - Supply shifts **LEFT**
   - **This is the key policy case — we'll analyze it in detail**

Example: An Increase in Saving



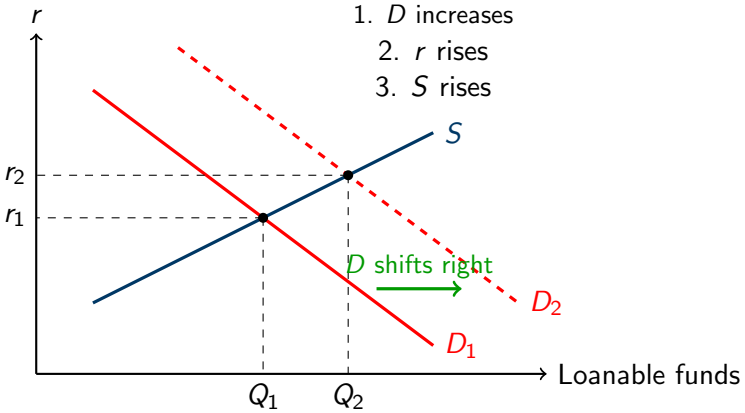
Result: More saving ⇒ Lower interest rate ⇒ More investment
More output is allocated to building future capital.

What Shifts the Demand Curve?

Recall: Demand = Investment (I)

1. **Technological innovation**
   - New technologies create profitable investment opportunities
   - Firms want to borrow more at every $r \Rightarrow$ Demand shifts **RIGHT**
2. **Investment tax credit**
   - Government gives tax breaks for buying capital
   - Investment becomes more attractive $\Rightarrow$ Demand shifts **RIGHT**
3. **Pessimistic expectations**
   - Firms expect lower future profits
   - Fewer projects look viable $\Rightarrow$ Demand shifts **LEFT**

Example: An Increase in Investment Demand



Result: More investment demand ⇒ Higher interest rate ⇒ More saving

The higher r incentivises households to consume less and save more.

The Consumption Function

So far we treated C as a given number. Now let's model household behaviour:

$$C = \bar{C} + c\,(Y - T) - e\,r$$

$\bar{C} > 0$ = autonomous consumption (basic needs)
c = **marginal propensity to consume** (MPC), $0 < c < 1$
$e \geq 0$ = sensitivity of consumption to the interest rate

Interpretation:

- c: if $c = 0.5$, households consume 50 cents of each extra euro of income
- e: higher r makes saving more attractive $\Rightarrow$ households consume less

Private saving follows immediately:

$$S_{priv} = (Y - T) - C = \underbrace{(1 - c)}_{\text{MPS}}(Y - T) - \bar{C} + e\,r$$

Saving depends **positively** on r (through e) — this is why the supply curve slopes upward, as we drew earlier.

The Investment Function

Similarly, let's formalise the demand curve we drew:

$$I = \bar{I} - d\,r$$

$$\begin{aligned} \bar{I} > 0 \quad &= \text{autonomous investment (baseline demand for capital)} \\ d > 0 \quad &= \text{sensitivity of investment to the interest rate} \\ r \quad &= \text{real interest rate} \end{aligned}$$

This is exactly our downward-sloping demand curve in algebraic form:

- Higher $r \Rightarrow$ lower I (fewer projects are profitable)
- The parameter d tells us *how sensitive* investment is to changes in r
- A large d means the demand curve is relatively flat (elastic)

Note: We are still in the **long run**. Output $\bar{Y}$ is fixed. These functions describe how that fixed output is **allocated**, not how much is produced.

Solving for Equilibrium Algebraically

With these functions, we can solve the loanable funds market:

Step 1: Compute national saving.

$$S = \underbrace{(1-c)(Y-T) - \bar{C} + er}_{S_{private}} + \underbrace{(T-G)}_{S_{public}}$$

Step 2: Set $S = I$ and solve for r.

$$(1-c)(Y-T) - \bar{C} + er + T - G = \bar{I} - dr$$

$$(d+e)\,r = \bar{I} + \bar{C} + G - (1-c)(Y-T) - T$$

$$\boxed{r^* = \frac{\bar{I} + \bar{C} + G - (1-c)(Y-T) - T}{d+e}}$$

Reading the parameters:

- e controls the **slope of the supply curve**: how much saving responds to r
- d controls the **slope of the demand curve**: how much investment responds to r
- The **numerator** captures shocks (ΔG, ΔT, changes in $\bar{C}$ or $\bar{I}$)

Numerical Example: Solving for r^*

Consider a closed economy in the long run:

$$\bar{Y} = 10{,}000 \quad T = G = 1{,}000 \quad C = 500 + 0.5(Y - T) - 100r \quad I = 6{,}000 - 400r$$

Step 1: Private saving depends on r.

$$S_{priv} = (Y - T) - C = 9{,}000 - 500 - 4{,}500 + 100r = 4{,}000 + 100r$$
$$S_{pub} = T - G = 0 \qquad S = 4{,}000 + 100r$$

Step 2: Set $S = I$ and solve for r.

$$4{,}000 + 100r = 6{,}000 - 400r \;\Rightarrow\; 500r = 2{,}000 \;\Rightarrow\; \boxed{r^* = 4\%} \quad I^* = 4{,}400$$

Both supply and demand respond to r: saving rises and investment falls as r increases, until they meet at $r^* = 4\%$ and $S^* = I^* = 4{,}400$.
The slope of S will matter for the **size of crowding out** — coming next.

What happens when government runs a deficit?

Recall (Topics 1-3): Public saving: $S_{public} = T - G$

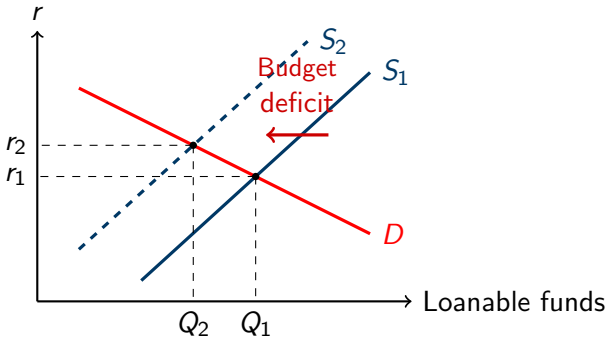
Budget deficit: $G > T \Rightarrow S_{public} < 0$

When government runs a deficit:
Public saving is **negative**, which **reduces** national saving: $S = S_{private} + \underbrace{S_{public}}_{<0}$

Example:

- $S_{private} = 200$ bn, deficit: $G - T = 50$ bn $\Rightarrow S_{public} = -50$
- National saving: $S = 200 + (-50) = 150$ bn
- The deficit absorbs 50 bn that could have gone to investment

Budget Deficit in the Loanable Funds Market



Step-by-step:

1. Deficit reduces national saving $\Rightarrow$ Supply shifts **LEFT**
2. Interest rate **RISES** ($r_1 \rightarrow r_2$)
3. Investment **FALLS** ($Q_1 \rightarrow Q_2$)

Less output is allocated to investment $\Rightarrow$ less future capital.

Crowding Out

Crowding Out

The decrease in private investment that results from government borrowing in the loanable funds market

The mechanism:

1. Government runs a deficit $\Rightarrow$ needs to borrow
2. Competes with firms for the available pool of saving
3. Pushes r up $\Rightarrow$ some investment projects become unprofitable
4. Private investment falls — it is "crowded out"

Why does this matter for growth?

- Less investment $\Rightarrow$ Less capital accumulation $\Rightarrow$ Lower future $\bar{Y}$

$\rightarrow$ Topic 3: Capital is crucial for long-run growth!

Numerical Example: Crowding Out

Using our economy ($C = 500 + 0.5(Y - T) - 100r$, $I = 6{,}000 - 400r$):

Initial: $T = G = 1{,}000 \Rightarrow S = 4{,}000 + 100r$, $r^* = 4\%$, $I^* = 4{,}400$.

Now G **rises by 200** (to 1,200), T unchanged:

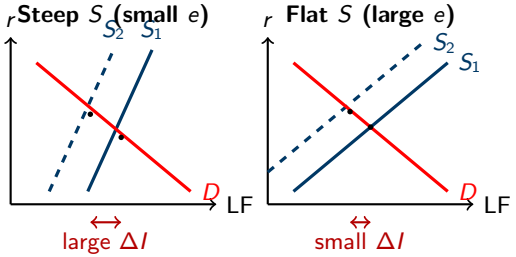
$$S_{pub} = 1{,}000 - 1{,}200 = -200 \qquad S_{priv} = 4{,}000 + 100r \text{ (unchanged)}$$
$$S = (4{,}000 + 100r) + (-200) = 3{,}800 + 100r$$

$$3{,}800 + 100r = 6{,}000 - 400r \quad \Rightarrow \quad r^* = 4.4\% \qquad I^* = 4{,}240$$

Crowding out = $\Delta I = 4{,}400 - 4{,}240 = 160 < 200$ — only **partial**.
Why not 200? When r rises, households consume less and save more.
The increase in private saving **partially offsets** the fall in public saving.

The Role of Slopes in Crowding Out

When a deficit shifts S left, the size of crowding out depends on the slopes:

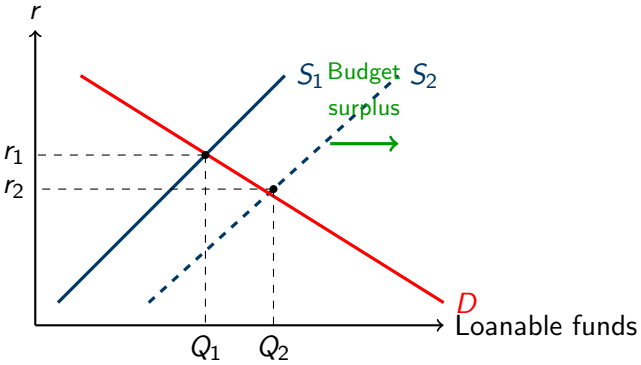


Supply slope (e): Flatter S (large e) $\Rightarrow$ private saving absorbs part of the deficit $\Rightarrow$ **less** crowding out. Steeper S (small e) $\Rightarrow$ saving cannot respond $\Rightarrow$ **more** crowding out.

Demand slope (d): Flatter D (large d) $\Rightarrow$ investment very sensitive to r, even a small Δr eliminates many projects $\Rightarrow$ **more** crowding out. Steeper D (small d) $\Rightarrow$ investment barely responds $\Rightarrow$ **less** crowding out.

Budget Surplus: The Opposite Effect

What if government runs a surplus? $T > G \Rightarrow S_{public} > 0$

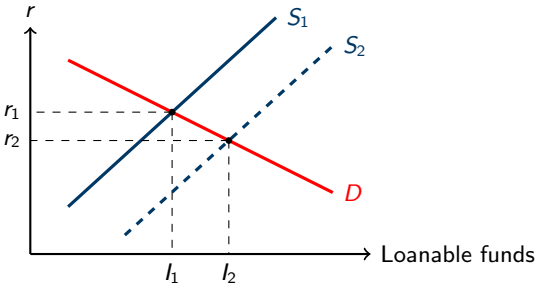


Budget surplus: Government adds to saving $\Rightarrow$ More funds available $\Rightarrow$ Lower r $\Rightarrow$ More investment

This is the mirror image of crowding out: the surplus "**crowds in**" private investment.

Policy: Tax Incentives for Saving

Policy: Reduce taxes on interest income. **Goal:** Encourage more saving.

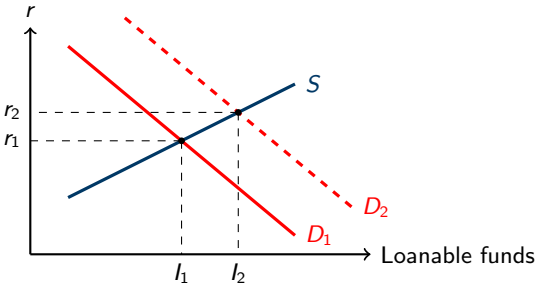


Effects:

- Tax break makes saving more attractive $\Rightarrow$ S shifts right
- Interest rate falls: $r_1 \rightarrow r_2$
- Investment rises: $I_1 \rightarrow I_2$
- More output allocated to capital accumulation

Policy: Investment Tax Credits

Policy: Tax breaks for firms buying capital. **Goal:** Encourage more investment.



Effects:

- Tax credit makes investment more profitable $\Rightarrow$ D shifts right
- r rises, which incentivises more saving (movement along S)
- Net effect: more investment and more saving

Summary: Key Concepts of Topic 4

1. **Allocation of Output (Long Run):**
   - $\bar{Y} = C + I + G$ — output is divided among three uses
   - Equilibrium OA = DA is equivalent to $S = I$
   - National saving: $S = Y - C - G = S_{private} + S_{public}$
2. **Financial System:**
   - Bonds, stocks, banks channel saving $\rightarrow$ investment
3. **Loanable Funds Market:**
   - Supply: saving (S). Demand: investment ($I = \bar{I} - dr$)
   - r adjusts to balance the intertemporal allocation
4. **Consumption, Saving & Investment Functions:**
   - $C = \bar{C} + c(Y - T) - er; \quad I = \bar{I} - dr$
   - e = slope of S; $\quad d$ = slope of D. Both affect crowding out size
5. **Fiscal Policy:**
   - Deficit $\Rightarrow$ Crowding out (size depends on slopes d and e)
   - Surplus $\Rightarrow$ Crowding in

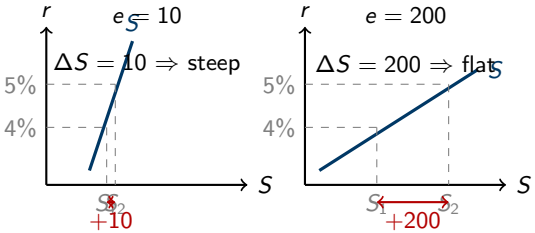
Appendix: Parameters vs Visual Slopes (1/2) — Supply

The confusion: e measures how much saving responds to r. Larger e = stronger response. So why does larger e make the supply curve **flatter**, not steeper?

The key: r is on the **vertical** axis, S on the **horizontal**. The visual slope is:

$$\text{visual slope} = \frac{\Delta r}{\Delta S} = \frac{1}{e} \qquad (\text{the } \textit{inverse} \text{ of } e)$$

Example: r rises by 1 percentage point. How much does saving move?



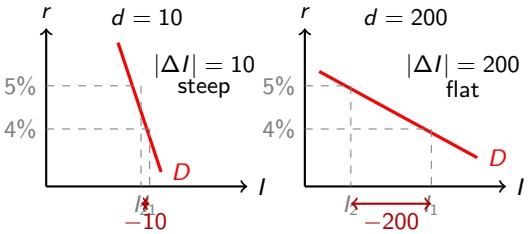
Both panels: same $\Delta r = 1$pp. With $e = 200$, saving jumps 200 to the right (the curve extends horizontally) ⇒ **flat**. With $e = 10$, saving barely moves ⇒ **steep**.

Appendix: Parameters vs Visual Slopes (2/2) — Demand

The same logic applies to the demand curve. From $I = \bar{I} - dr$:

$$\text{visual slope of } D = \frac{\Delta r}{\Delta I} = \frac{1}{d}$$

Example: r rises by 1 percentage point. How much does investment fall?



Rule of thumb: the sensitivity parameter (e or d) measures the *horizontal* response to a change in r (vertical). A larger parameter means more horizontal movement per unit of r $\Rightarrow$ **flatter curve**. The visual slope is always 1/parameter.